

Because ZFS allocates its metadata as it needs it, no initial space cost is required, and the number of files is limited only by the available space. The output from the `df -g` command must be interpreted differently for ZFS than other file systems. The total files reported is only an estimate based on the amount of storage that is available in the pool. ZFS is a transactional file system. Most file system modifications are bundled into transaction groups and committed to disk asynchronously. Until these modifications are committed to disk, they are termed pending changes. The amount of space used, available, and referenced by a file or file system does not consider pending changes. Pending changes are generally accounted for within a few seconds. Even committing a change to disk by using `fsync(3c)` or `O_SYNC` does not necessarily guarantee that the space usage information is updated immediately.

4.8.6 Mounting ZFS File Systems

ZFS is designed to reduce complexity and ease administration. For example, with existing file systems you must edit the `/etc/vfstab` file every time you add a new file system. ZFS has eliminated this requirement by automatically mounting and unmounting file systems according to the properties of the dataset. You do not need to manage ZFS entries in the `/etc/vfstab` file.

4.9 ZFS Snapshots and Clones

By now, you already have an overview of ZFS snapshots. In this section, we will learn some commands to use snapshots and clones.

4.9.1 Creating and destroying ZFS snapshots

Snapshots are created by using the `zfs snapshot` command, which takes as its only argument the name of the snapshot to create. The snapshot name is specified as follows:

```
filesystem@snapname
```

```
volume@snapname
```

In the following example, a snapshot of `tank/home/ahrens` that is named `friday` is created.

```
# zfs snapshot tank/home/ahrens@friday
```

You can create snapshots for all descendent file systems by using the `-r` option. For example:

```
# zfs snapshot -r tank/home@now
```

```
# zfs list -t snapshot
```